ENVIRONMENTAL REVIEW · COMPREHENSIVE EXAM

Fifty years of cooling cities: what the evidence actually supports

A literature review of urban heat-island mitigation, from shade trees to reflective roofs.

The field has converged on the question, but not on the answer.

Cities run several degrees hotter than the countryside around them, and the gap is widening with climate change. Fifty years of research agrees the urban heat island is real and harmful — but disagrees sharply on which interventions cool a city most per dollar.

What I reviewed, and how I selected it.

I screened 1,140 studies and retained 86 that measured a temperature outcome from a built intervention with a credible control. The corpus spans 1974 to 2025 and four intervention families: urban trees, reflective surfaces, green roofs, and water features.

- 86 studies retained from 1,140 screened.
- Inclusion required a measured temperature outcome and a control.

HOW THE FIELD EVOLVED

Five decades, four shifts in what the field measured.

1974

Heat island documented

Early studies establish that cities are measurably hotter than rural surroundings.

1990

Albedo era

live

Research pivots to reflective roofs and pavements as the first engineered fix.

2005

Green infrastructure

Urban trees and green roofs reframed as multi-benefit cooling.

2018

Equity turn

decision

Studies show heat burden falls hardest on low-income neighbourhoods.

2024

Whole-system models

at-risk

Coupled climate-energy models question whether single fixes scale citywide.

The early consensus — reflect more sunlight — was right but incomplete.

The albedo era proved reflective roofs cut surface temperatures sharply and cheaply. But later work showed the cooling is local and daytime-only, and that wholesale reflective pavement can push glare and heat onto pedestrians. The fix worked; its side effects were under-counted.

- Reflective surfaces cool roofs reliably and cheaply.
- Benefits are local, daytime-only, and can displace heat to the street.

Four mitigation families, compared on what the evidence shows.

CRITERION	URBAN TREES	REFLECTIVE ROOFS	GREEN ROOFS	WATER FEATURES
Peak cooling	2–4°C	1–3°C	1–2°C	1–2°C
Cost per m²	Low	Lowest	Highest	Medium
Co-benefits	High	Low	High	Medium
Evidence base	Strong	Strong	Moderate	Weak
Equity reach	Uneven	Broad	Narrow	Narrow

Where the literature genuinely agrees: trees do the most, if they survive.

Across 31 tree studies, mature canopy delivers the largest and most durable cooling — 2 to 4 degrees — plus air-quality and stormwater co-benefits no other intervention matches. The catch is establishment: young trees in hot, paved sites die before they shade anything.

- Mature canopy is the clear leader on cooling and co-benefits.
- Survival, not planting, is the binding constraint.

Where it disagrees: nobody has cleanly compared interventions at city scale.

Almost every study tests one intervention at one site. The handful of citywide model comparisons rely on assumptions that drive their own conclusions. We have strong evidence for each fix in isolation and almost none for the right *mix* across a real city.

- Single-site, single-intervention designs dominate the corpus.
- Citywide comparative evidence is the field's largest gap.

The equity gap is a measurement gap, not just a policy one.

The hottest neighbourhoods are the least studied — most temperature sensors sit in wealthier, greener districts. Until monitoring follows the heat burden rather than the budget, equity claims rest on models, not measurements. This is where I argue the next decade of work belongs.

- Sensor coverage tracks wealth, not heat exposure.
- Equity findings currently lean on modelled, not measured, data.

Synthesis: the field needs comparative, equity-weighted, whole-city trials.

The evidence supports a portfolio — trees where they survive, reflective surfaces where they don't, green roofs for co-benefits — but no study has tested that portfolio against the real distribution of urban heat. That comparative, equity-weighted trial is the contribution the field is missing.

- A portfolio, not a single silver-bullet fix.
- Test it against the actual map of who bears the heat.

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We know what cools a building — we still don't know what cools a city fairly.